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# Grasping mode analysis and adaptive impedance control for apple harvesting robotic grippers

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ABSTRACT

In order to reduce the mechanical damage caused by apple harvesting robot to apples and realize the non-destructive grasping during the grasping process, an adaptive impedance control strategy for apple harvesting robot compliant grasping is proposed based on the analysis of grasping modes in this paper. Firstly, burgers viscoelastic model is established to describe the rheological characteristics of apples. Then taking the plastic deformation of apples as the damage standard, three types of grasping modes of the gripper for picking apples are compared and analyzed. Finally, in view of the fact that the traditional impedance controller cannot satisfy the grasping force control under the condition of uncertain environmental stiffness and uncertain environmental position, an adaptive controller is designed to adjust the desired position online based on the error between desired force and actual grasping force so that it can track the desired force. Simulation and experiment results show that the adaptive impedance control strategy is more compliant with the actual grasping force, the overshoot and the adjustment time has been significantly improved, which provides a reference for actual apple picking.

## 1. Introduction

The stable and compliant grasping of the apple harvesting robot is one of the key steps to realize the intelligent harvesting operation (Chen et al., 2019; Peng et al., 2017; Ji et al., 2020b), it can be achieved by selecting the optimal grasping mode and precisely controlling the grasping force of the gripper. However, owing to the different biological characteristics of the target apple, there are often situations in which apples fail to be grasped result from the inability to precisely control the grasping force during the picking process. If the grasping force is too large, it will cause irreversible mechanical damage to the target apple and reduce the quality of the apple. While, the grasping force is too small, it will cause the apple to slide or even fall off, resulting in failure to grasp. Therefore, studying the non-destructive grasping technology of the apple harvesting robot is of great significance for reducing the mechanical damage to the apple and improving the success rate of apple picking (Zhao et al., 2016).

In order to solve existing problems in compliant grasping, a lot of research in the field of robotic harvesting for various fruit crops have been conducted by scholars. For example, Maxwell model and burgers model were established to describe the anisotropic stress relaxation behavior and creep characteristics of the apple respectively and then the accuracy of the model was verified by creep experiments on two different kinds of Iranian apples (Gorji Chakespari et al., 2010). Based on this work, the biological characteristics parameters of the apple were measured by compression test, and then a finite element model was established to test the stress and deformation of each part of the apple under different grasping speeds to determine the optimal grasping force (Ji et al., 2019). Some scholars carried out the damage analysis by establishing a finite element model to analyze the viscoelastic behavior of the contact between the apple and the plane and the contact between the apple and the apple (Ahmadi et al., 2016). Moreover, an apple compression static load response model which similar to burgers model was established and the relationship between stress relaxation duration and deformation of apple was verified through a compression test (Casandroiu et al., 2014). Elien et al. (2016) established a viscoelastic-plastic contact force model (VEP) for apples based on the elastoplastic Thornton model, and the relationship between contact force and apple deformation was verified through half-fruit compression tests. There were some other studies, such as the grasping model of the end effector

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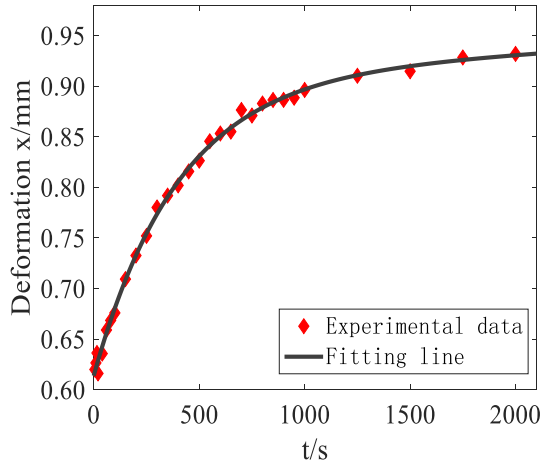



**Fig. 1.** Apple deformation fitting curve.

and the target fruit was equivalent to the second-order impedance-admittance model, and the reference position was adjusted by the force-based impedance control strategy so that the grasping force could meet the control requirements (Wang et al., 2015). In order to avoid errors caused by uncertain environmental information, the relationship between the force and deformation of the apple was equivalent to an environmental function and the impedance controller was self-tuning to make the contact force smoother (Ji et al., 2020a). Zhou et al. (2014) used neural fuzzy network combined with gradient descent method to effectively limit the force overshoot and reduce the mechanical damage suffered by fruits and vegetables. Xu (2015) proposed an impedance control algorithm that allows the gripper to smoothly transition from free motion to constrained motion. Stachowsky et al. (2016) studied a precise grasping strategy by combining force control and sliding detection. According to the optimal control theory, the contact collision model was improved and optimized and a force tracking impedance control strategy was proposed (Roveda et al., 2016).

To summarize, although the above-mentioned research carried out a comprehensive analysis on different aspect to grasping force control, it did not consider the influence of different grasping modes, variable environmental stiffness and variable environmental position on apple picking. In view of this, the objective of this study is to improve the traditional impedance control system for apple harvesting robot to realize compliant picking. The study takes the plastic deformation of apples as the damage standard and conducts damage analysis on three different grasping modes during the grasping process. On the basis, the adaptive impedance control system is designed to make the gripper quickly and stably grasp apples with low overshoot even when the environmental stiffness and environmental position are not clear. Finally, the effectiveness of proposed method is verified by the comparison of simulation and experiments.

## 2. Materials and methods

### 2.1. Apple viscoelastic parameter acquisition

Apple is a kind of crop with viscoelastic properties. When it is loaded, it will flow and deform. In order to accurately describe the viscoelastic properties of apples, burgers viscoelastic model is adopted in this paper. The creep equation is shown in Eq. (1):

$$\ddot{f}(t)+\left(\frac{e_0}{\eta_0}+\frac{e_1}{\eta_1}+\frac{e_0}{\eta_1}\right)\dot{f}(t)+\frac{e_0e_1}{\eta_0\eta_1}f(t)=e_0\ddot{x}(t)+\frac{e_0e_1}{\eta_1}\dot{x}(t) \tag{1}$$

Among them, $e_0$ is the instantaneous elastic coefficient, $e_1$ is the delayed elastic coefficient, $\eta_0$ is the viscosity coefficient of series viscous elements, $\eta_1$ is the viscosity coefficient of parallel viscous elements, the constant load compression test of apple was carried out to obtain the viscoelastic coefficient of apple, and the relationship between the deformation of apple and time was shown in Fig. 1.

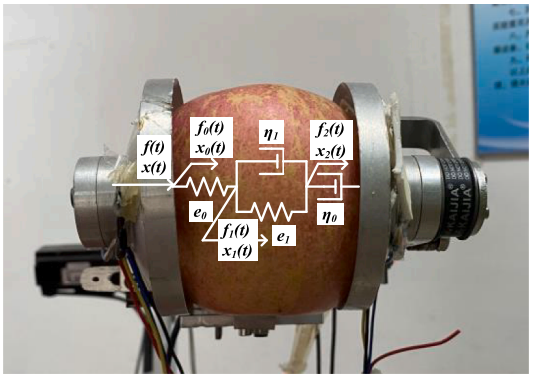



**Fig. 2.** Apple stress analysis.

According to the fitting results, the viscoelastic parameters of apple were obtained as:

$$\begin{cases} e_0 = 24.43\ N{\cdot}mm^{-1} \\ e_1 = 50.88\ N{\cdot}mm^{-1} \\ \eta_0 = 1.27\times10^{6}\ N{\cdot}S{\cdot}mm^{-1} \\ \eta_1 = 2.03\times10^{4}\ N{\cdot}S{\cdot}mm^{-1} \end{cases} \tag{2}$$

### 2.2. Grasping modes analysis

#### 2.2.1. Apple stress analysis and plastic model establishment

When the target apple is grasped, the stress diagram of each deformation layer is shown in Fig. 2. Where $f(t)$ and $x(t)$ represent the contact force and total deformation of the apple, $f_0(t)$, $f_1(t)$ and $f_2(t)$ represent the component forces on the elastic unit layer and viscous unit layer of the apple respectively, and $x_0(t)$, $x_1(t)$ and $x_2(t)$ describe the deformation of the elastic unit layer and viscous unit layer of the apple respectively. The mathematical relationship between them are shown in Eqs. (3)–(8):

$$x(t)=x_0(t) \tag{3}$$

$$f(t)=f_0(t)=f_1(t)=f_2(t) \tag{4}$$

$$f_0(t)=e_0(x_0(t)-x_1(t)) \tag{5}$$

$$f_1(t)=e_1(x_1(t)-x_2(t))+\eta_1\left(\dot{x}_1(t)-\dot{x}_2(t)\right) \tag{6}$$

$$f_2(t)=\eta_0\dot{x}_2(t) \tag{7}$$

$$x(t)=x_i-f(t)/k_i \tag{8}$$

where $x_i$ and $k_i$ are the displacement and stiffness of the gripper, integrating the above equation, the relationship between the grasping force and the displacement of the gripper is obtained:

$$\frac{\eta_0\eta_1(e_0+k_i)}{e_0e_1k_i}\ddot{f}(t)+\frac{k_i(\eta_0e_0+\eta_1e_0+\eta_0e_1)+\eta_0e_0e_1}{e_0e_1k_i}\dot{f}(t)+f(t) =\frac{\eta_0\eta_1}{e_1}\ddot{x}_i(t)+\eta_0\dot{x}_i(t) \tag{9}$$

According to the above analysis, the plastic deformation of apple mainly comes from the viscous components, so this paper defines the mathematical model of apple plastic deformation as:

$$x_2(t)=\frac{1}{\eta_0}\int_0^T f(t)dt \tag{10}$$

where $T$ is stable grasping time, which refers to the time taken by the

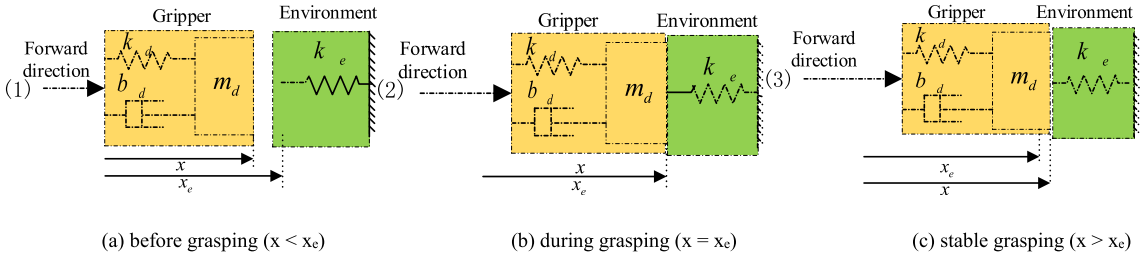



**Fig. 3.** Grasping model between gripper and apple for impedance control.

gripper from the moment it touches the apple to the moment it stably grasps the apple. Combining the viscoelastic coefficient of apple obtained from the above apple compression experiment and the defined plastic deformation model, a theoretical analysis for the apple plastic deformation caused by grasping force was carried out.

### 2.2.2. Analysis and verification of damage to apple caused by three grasping modes

The speed of the gripper gradually changed from a certain initial speed to zero during the process that the gripper touched apple until it stably grasped apple (Zhang et al., 2018a). The grasping modes were categorized into three types based on the speed change trend: (1) the two-finger gripper closed at a prescribed speed and then uniformly decelerated to zero after contacting the target apple; (2) the two-finger gripper closed at a prescribed speed after contacting the target apple and subsequently decelerated to zero; (3) the two-finger gripper closed at a prescribed speed and then decelerated variably to zero after contacting the target apple.

$$\begin{cases} v_1(t) = v_0 - v_0 t/T \\ v_2(t) = v_0/\sqrt{1+(2t/T)^8} \\ v_3(t) = v_0\left(0.01T^2\right)^t \end{cases} \tag{11}$$

We utilized the change trend of grasping speed expressed by Eq. (11) to describe the three grasping modes mentioned above. And then the three grasping modes were substituted into the mathematical model represented by Eq. (9). Finally, the change trend of the grasping force and the apple plastic deformation generated by three different grasping modes were obtained through the numerical calculation and analysis.

## 2.3. Adaptive impedance control for compliant grasping of apple picking robot

The impedance control strategy is widely used in robot compliance control due to its strong robustness. However, in the actual grasping process of apple harvesting robot, the control performance of traditional impedance control is not satisfactory due to the uncertainty of environmental stiffness and environmental position (Zhang et al., 2018b). In this section, on the basis of position-based impedance control strategy (Hogan, 1985), an adaptive impedance control strategy is proposed for the one DOF two-finger gripper of the designed apple picking robot to adjust the actual grasping force under the condition of fixed environmental stiffness and environmental position and the condition of variable environmental stiffness and environmental position.

### 2.3.1. Position-based impedance control strategy

The position-based impedance control strategy depends on the combined model of damping spring mass system and first-order spring system. The first-order spring model is used to simulate the apple with complex biological characteristics. Fig. 3 describes the three stages of the gripper and the apple from never contacting ($x < x_e$) to stably grasping ($x > x_e$). When the grasping process is in an ideal state (the environmental stiffness and the environmental position remain unchanged), it means that the biological characteristics of the apple are not changed or the gripper does not slide. Otherwise, it means that the control precision of grasping force is not enough due to some other reasons, for example, inappropriate grasping point. Taking the one DOF two-finger gripper as an example, the model of impedance controller is shown in Eq. (12):

$$m_d\left(\ddot{x}_r - \ddot{x}\right) + b_d\left(\dot{x}_r - \dot{x}\right) + k_d(x_r - x) = F_r - k_e(x - x_e) \tag{12}$$

where $x_r$ and $x$ represent the desired position and actual position of the gripper, $F_r$ represent the desired force, $k_e$ and $x_e$ represent environmental stiffness and environmental position, $m_d$, $b_d$ and $k_d$ represent the inertia parameters, damping parameters and stiffness parameters of the impedance controller, respectively.

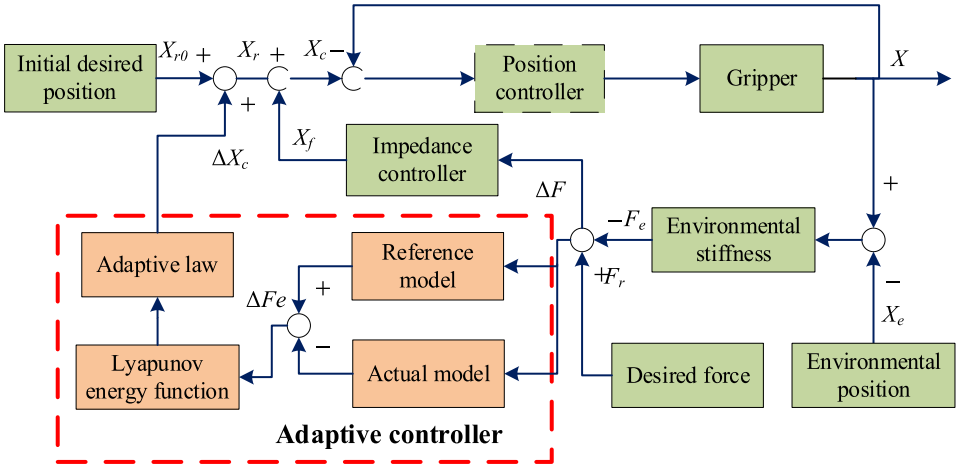



**Fig. 4.** The Adaptive impedance control block diagram for gripper.

*2.3.2. Adaptive impedance control strategy*

Aiming at the uncertainty of grasping environmental information (Xu and Sun, 2020; Zhang et al., 2020; Zhang and Khamesee, 2017), this paper improves the traditional impedance control system by adding a small position adjustment $\Delta x_c$ to the desired position $x_{r0}$ to adjust the actual grasping force so that it can realize compliant grasping. The adaptive impedance control block diagram is shown in Fig. 4.

For the one DOF two-finger gripper in this paper, to achieve the purpose of adaptive control, the new desired position can be expressed as follows:

$$x_r = x_{r0} + \Delta x_c \tag{13}$$

where $x_{r0}$ is the initial desired position and $\Delta x_c$ is the position adjustment generated by the adaptive controller. Taking $\Delta x_c$ as follows:

$$\Delta x_c = a(t) + c(t)\Delta F(t) + d(t)\Delta \dot{F}(t) \tag{14}$$

where $\Delta F(t)$ is the error between the desired force and the contact force; $a(t)$ is the auxiliary term, $c(t)$ and $d(t)$ are the proportion and differential factor of the adaptive controller. By combining Eqs. (12), (13) and (14), a new force error balance equation is obtained:

$$\begin{aligned}\Delta \ddot{F} + \frac{b_d + k_d k_e d(t)}{m_d}\Delta \dot{F} + \frac{k_d - k_e + k_d k_e c(t)}{m_d}\Delta F \\ = \frac{k_d F_r + k_d k_e x_e - k_d k_e x_{r0} - k_d k_e a(t)}{m_d}\end{aligned} \tag{15}$$

According to the above force error equation, the following result can be obtained:

$$\Delta \ddot{F} + A_p(t)\Delta \dot{F} + B_p(t)\Delta F = R(t) \tag{16}$$

Among them:

$$\begin{cases} A_p(t) = \dfrac{b_d + k_d k_e d(t)}{m_d} \\ B_p(t) = \dfrac{k_d - k_e + k_d k_e c(t)}{m_d} \\ R(t) = \dfrac{k_d F_r + k_d k_e x_e - k_d k_e x_{r0} - k_d k_e a(t)}{m_d} \end{cases} \tag{17}$$

Take $\Delta f_p = \begin{bmatrix} \Delta F \\ \Delta \dot{F} \end{bmatrix}$, the simplified force error equation can be rewritten as follows:

$$\Delta \dot{f}_p = \begin{bmatrix} 0 & 1 \\ -B_p(t) & -A_p(t) \end{bmatrix}\Delta f_P + \begin{bmatrix} 0 \\ R(t) \end{bmatrix} = a_p(t)\Delta f_p + \begin{bmatrix} 0 \\ R(t) \end{bmatrix} \tag{18}$$

In order to make the actual output reach a more ideal state, the selected reference model is an ideal second-order system:

$$\Delta \ddot{F}_m + A_m \Delta \dot{F}_m + B_m \Delta F_m = 0 \tag{19}$$

Take $\Delta f_m = \begin{bmatrix} \Delta F_m \\ \Delta \dot{F}_m \end{bmatrix}$, and rewrite the above reference model as follows:

$$\Delta \dot{f}_m = \begin{bmatrix} 0 & 1 \\ -B_m & -A_m \end{bmatrix}\Delta f_m = a_m(t)\Delta f_m \tag{20}$$

By subtracting Eq. (20) from Eq. (18), the error between the reference model and actual model can be obtained:

$$\begin{aligned}\Delta \dot{F}_e &= a_m \Delta F_e + \left(a_m - a_p(t)\right)\Delta f_p - \begin{bmatrix} 0 \\ R(t) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ B_p(t) - B_m & A_p(t) - A_m \end{bmatrix}\begin{bmatrix} \Delta F \\ \Delta \dot{F} \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 & 1 \\ -B_m & -A_m \end{bmatrix}\Delta F_e + \begin{bmatrix} 0 \\ -R(t) \end{bmatrix}\end{aligned} \tag{21}$$

where $\Delta F_e = \begin{bmatrix} \Delta F_m - \Delta F \\ \Delta \dot{F}_m - \Delta \dot{F} \end{bmatrix}$, and the quadratic energy function constructed based on the Lyapunov stability theorem II can be written as follows:

$$\begin{aligned}V(\Delta F_e, t) &= \frac{1}{2}\Delta F_e^{\mathrm{T}} P \Delta F_e + \frac{1}{2} Z^{\mathrm{T}} R Z \\ &= \frac{1}{2}\Delta F_e^{\mathrm{T}} P \Delta F_e + \frac{1}{2}\alpha_0 \left(B_p(t) - B_m\right)^2 \\ &\quad + \frac{1}{2}\alpha_1 \left(A_p(t) - A_m\right)^2 + \frac{1}{2}\alpha_2 R(t)^2\end{aligned} \tag{22}$$

where $\alpha_0$, $\alpha_1$, and $\alpha_2$ are positive constants, $P$ is non-singular positive definite matrices.

$$Z = \begin{bmatrix} B_p(t) - B_m \\ A_p(t) - A_m \\ R(t) \end{bmatrix} P = \begin{bmatrix} P_1 & P_2 \\ P_2 & P_3 \end{bmatrix}$$

According to Lyapunov stability theorem II, as long as there exists a positive definite symmetric matrices $Q$ and satisfies $a_m^{\mathrm{T}} P + P a_m = -Q$, then there exists $P$ in above equation. By differentiating the quadratic energy function, the following result can be obtained:

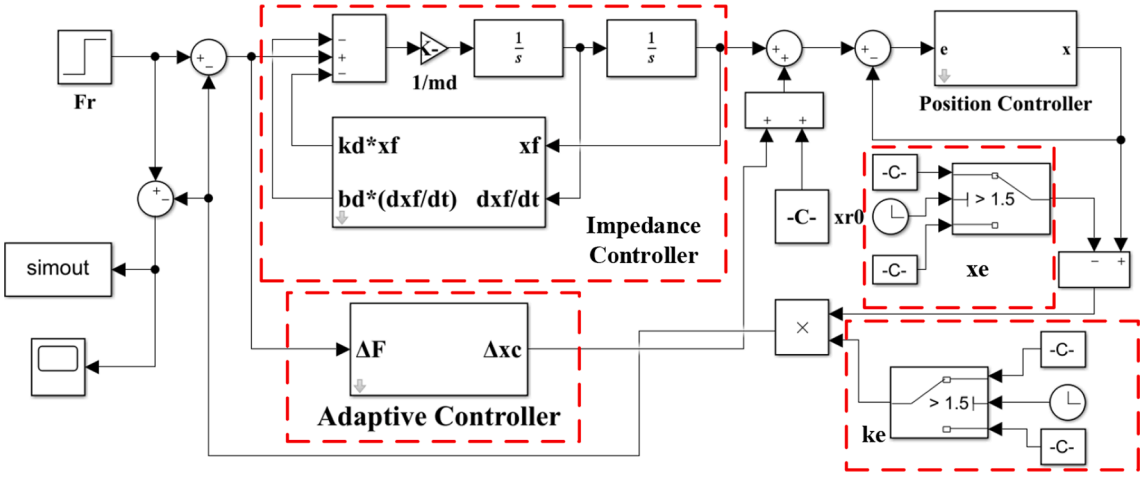
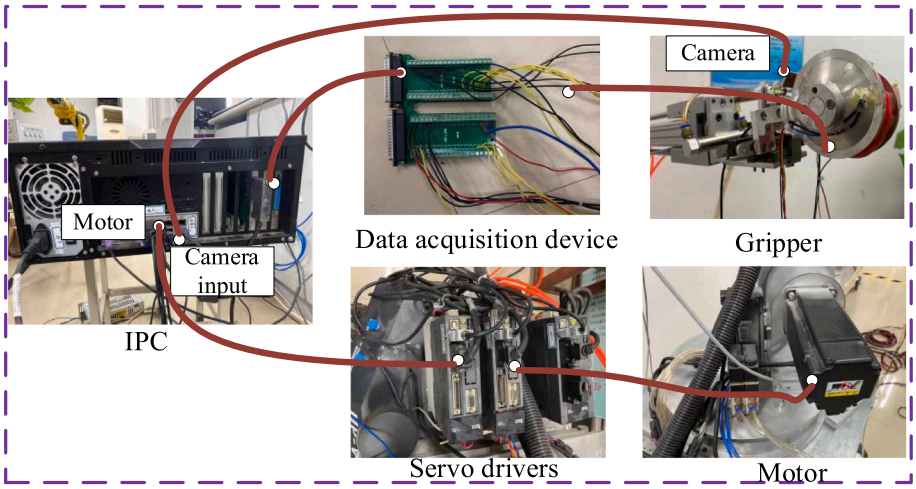



**Fig. 5.** Simulation model of adaptive impedance control system.



(a) Experimental structure of apple harvesting robot




(b) Machine drive and data transmission equipment

**Fig. 6.** Apple harvesting robot experiment platform.

$$\begin{aligned}\dot{V}(\Delta F_e,t)&=\left(B_p(t)-B_m\right)\left(\chi\Delta F+\alpha_0\dot{B}_p(t)\right)\\&+\left(A_p(t)-A_m\right)\left(\chi\Delta\dot{F}+\alpha_1\dot{A}_p(t)\right)\\&+R(t)\left(\alpha_2\dot{R}(t)-\chi\right)-\frac{1}{2}\Delta F_e^{\mathrm{T}}Q\Delta F_e\end{aligned}\tag{23}$$

where $\chi=P_2(\Delta F_m-\Delta F)+P_3\left(\Delta\dot{F}_m-\Delta\dot{F}\right)$, in this case, it is only necessary to guarantee $\dot{V}(\Delta F_e,t)<0$ that the stability condition can be satisfied. Hence take the first three terms of Eq. (23) as 0, and the following equation can be derived:

$$\begin{cases}\dot{A}_p(t)=-\dfrac{\chi}{\alpha_1}\Delta\dot{F}\\ \dot{B}_p(t)=-\dfrac{\chi}{\alpha_0}\Delta F\\ \dot{R}(t)=\dfrac{\chi}{\alpha_2}\end{cases}\tag{24}$$

Thus, the control law of the adaptive controller can be obtained:

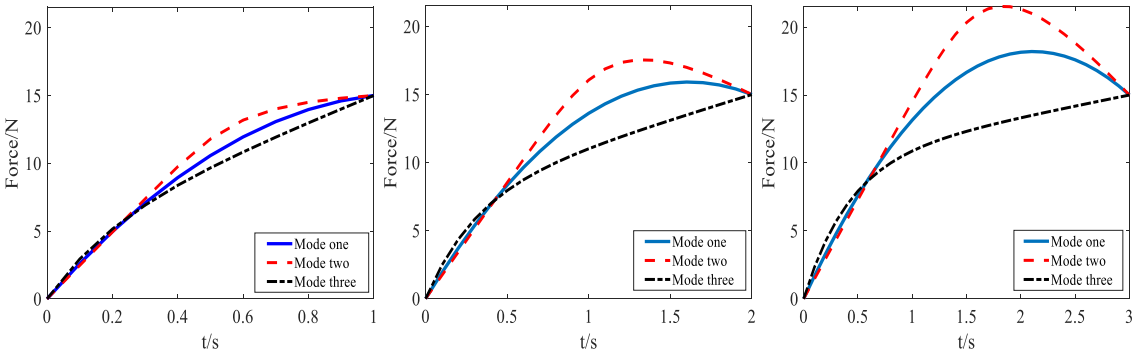
$$\begin{cases}a(t)=a_0-\delta_0\int_0^t\chi dt\\ c(t)=c_0-\delta_1\int_0^t\chi\Delta F dt\\ d(t)=d_0-\delta_2\int_0^t\chi\Delta\dot{F}(t)dt\\ \chi(t)=-\left(\eta_p\Delta F(t)+\eta_v\Delta\dot{F}(t)\right)\end{cases}\tag{25}$$

where $\delta_0$, $\delta_1$, $\delta_2$, $\eta_p$ and $\eta_v$ are small positive constants, and the control requirements of the actual contact force can be satisfied by selecting the appropriate coefficient value.

### 2.4. Simulation and verification design for adaptive impedance control strategy

A simulation model of the one DOF two-finger gripper was designed to verify the feasibility of the adaptive impedance control strategy, as shown in Fig. 5. Step force and sine force tracking experiments were carried out under the condition of fixed environmental stiffness and environmental position, variable environmental stiffness and variable environmental position. Finally, the force control performance of traditional impedance control strategy and adaptive impedance control strategy were compared and analyzed.

The above simulation model made apple equivalent to a first-order




(a) T=1s (b) T=2s (c) T=3s

**Fig. 7.** Comparison of the grasping force under three grasping modes.

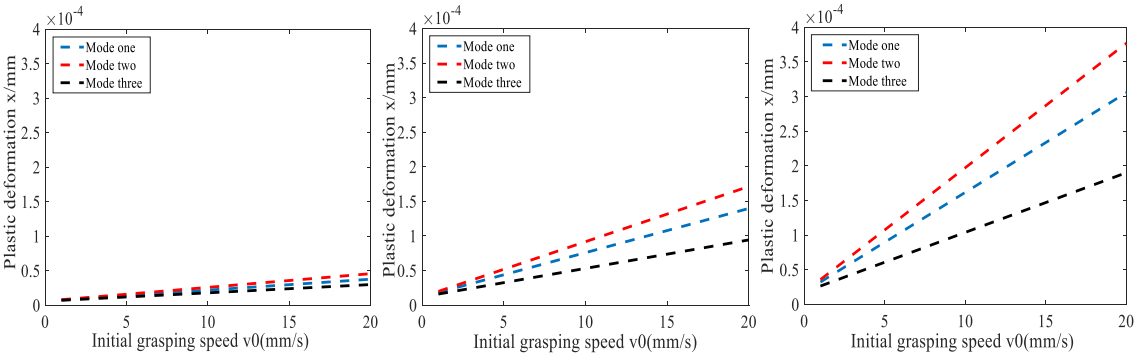
spring model, and the actual grasping force of the apple $F_e$ was obtained through the environmental stiffness $k_e$, the environmental position $x_e$ and the actual position of the gripper x. Then the error between the actual grasping force $F_e$ and the desired force $F_r$ was taken as the input of the adaptive controller, meanwhile the adaptive controller would generate a position adjustment $\Delta x_c$ which was taken as the input of the position controller together with the output of the impedance controller and the actual position of the gripper, so that the actual position of the gripper could be quickly adjusted to the optimal position, and the actual grasping force of the gripper could always be within a small error range. Finally, we observed the force tracking results through the oscilloscope.

In this simulation model, the value of the module representing the environmental stiffness and environmental position was set to the same value before and after 1.5 s, which was used to simulate the situation that the environmental stiffness and environmental position did not change (that is, biological characteristics of apple did not change or the gripper did not slide during grasping process). Changing one set of values at 1.5 s was used to simulate the situation where the environmental stiffness or environmental position changed. so that, we could observe the control performance of the adaptive impedance strategy. The impedance controller was constructed based on the position-based impedance control theory, by selecting the appropriate inertia parameter ($m_d$), damping parameter ($b_d$) and stiffness parameter ($k_d$), the error between the desired position and the actual position and its differential term were adjusted, so that the actual grasping force and the desired force of the gripper were kept within a certain range. The adaptive controller constructed according to the designed adaptive algorithm and the result of control law in the above chapter was to prevent the apple from damaging or slipping when the force error was too large. The encapsulated model and adjustment parameters like $\delta_0$, $\delta_1$ $\delta_2$ were set to appropriate values, and when the environmental stiffness or environmental position changed, the adaptive controller worked and adjusted actual grasping force of the gripper rapidly.

### 2.5. Apple harvesting robot experiment setup

The whole apple harvesting experimental platform is shown in Fig. 6. It is mainly composed of moving car, manipulator, gripper with sensor system, visual recognition system and control system and so on. The 5 DOF manipulator with PRRRP structure is optimized to simplify the control strategy. The one DOF two-finger gripper is designed to meet the requirements of the compliant picking for apple harvesting robot. The sensor system mainly includes CCD camera, infrared sensors and FSR force sensors to complete the acquisition of apple image, detection of whether the apple is within the grasping range and the detection of grasping force value. When the apple harvesting robot started grasping task, the CCD camera firstly obtained the image of the apple and it was converted into a digital signal to be transmitted to the visual recognition system. Then the position information of the target apple would be sent to the control system and the control system started to work, which drove the gripper to the designated position. Finally, when the infrared sensor was triggered, the gripper started to grasp the apple.

In order to verify the effectiveness of the adaptive impedance control strategy, in addition to the above verification of the simulation model to grasp apples, a comparative experiment between the traditional impedance control strategy and the adaptive impedance control strategy was conducted through the experimental equipment shown in Fig. 6. We controlled the motor to drive the gripper through the upper computer software, and limit the maximum output current and torque of the motor. The normal apples with diameter of 70–80 mm and mass of 0.18–0.22 kg were selected for grasping. At the same time, in order to avoid the contingency of the experiment, we did 10 groups of grasping




(a) T=1s (b) T=2s (c) T=3s

**Fig. 8.** Comparison of the apple plastic deformation under three grasping modes.

**Table 1**
Parameters of adaptive impedance controller.

| Parameter | $m_d$ | $b_d$ | $k_d$ | $\eta_p$ | $\eta_v$ | $\delta_0$ | $\delta_1$ | $\delta_2$ | $x_e$ | $k_e$ | $F_r$ | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | 1 | 100 | 4000 | 0.05 | 0.001 | 0.0001 | 0.1 | 0.1 | 0.004 | 15,000 | 8 | sin(5 t) + 8 |

experiments on apple to ensure the stability of proposed adaptive impedance control algorithm. In the process of completing the grasping task, the grasping force information between the gripper and apple would be detected by the FSR force sensor and the data would be gathered by a data acquisition card to transmit to IPC.

## 3. Result and discussion

### 3.1. Analysis of simulation results for different grasping modes

In order to ensure that the gripper could grasp apple stably, the force value was 15 N, the initial grasping speed was 1 mm/s, and the stable grasping time was 1 s, 2 s and 3 s respectively.

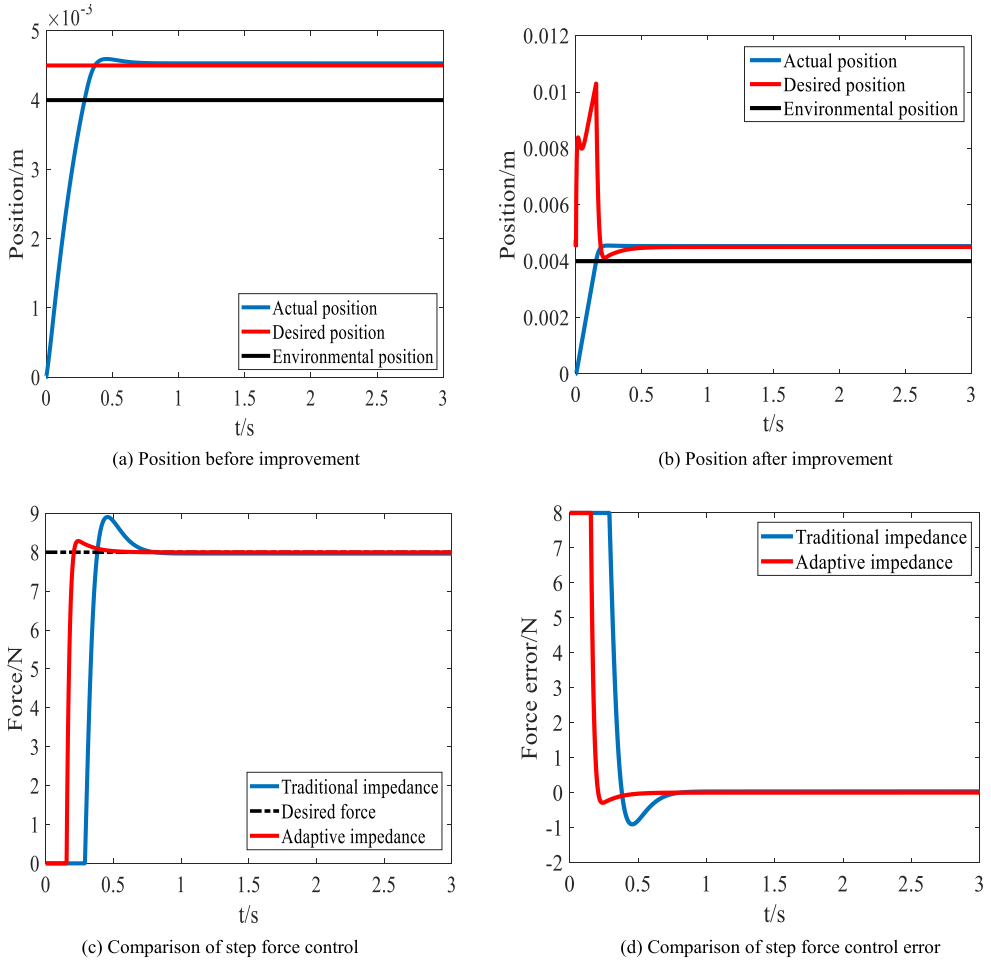
As shown in Fig. 7, with the increase of stable grasping time, the increasing trend of grasping force generated by three grasping modes tended to be distinctive. The third grasping mode produced the smallest grasping force compared with the first two modes until the completion of the stable grasping action.

The results of the change in grasping force produced by three grasping modes were substituted into the plastic deformation model defined in Eq. (10) for calculation, and the plastic deformation was calculated from the initial grasping speed 1 mm/s to 20 mm/s respectively. The variation of plastic deformation with initial grasping speed under different stable grasping time was reflected in Fig. 8.

According to variation trend of the apple plastic deformation shown in Fig. 8, there were obvious differences in the growth of plastic deformation caused by the three grasping modes with the increase of initial grasping speed, and the plastic deformation produced by the third grasping mode was smallest. At the same time, when the initial grasping speed was 1 mm/s and the grasping time was 1 s, 2 s, and 3 s, the plastic deformation produced by the third grasping mode was the smallest which was consistent with the law of the grasping force reflected in Fig. 7.

The basic reason why the third grasping mode produces the least damage to the apple is that when the gripper grasps the apple at a certain initial speed, the first and the second grasping modes make the gripper close faster than the third grasping mode in the process of touching the apple to stably grasping the apple, so under the same stable grasping time, the closing distance of (final actual position) the gripper in the first two grasping modes is larger than that in the third one, which leads to the first two grasping modes produce more grasping force than the third grasping mode. As a result, the damage caused by the third mode is also lower than that caused by the first two modes.

Therefore, in order to reduce the irreversible damage to apple and



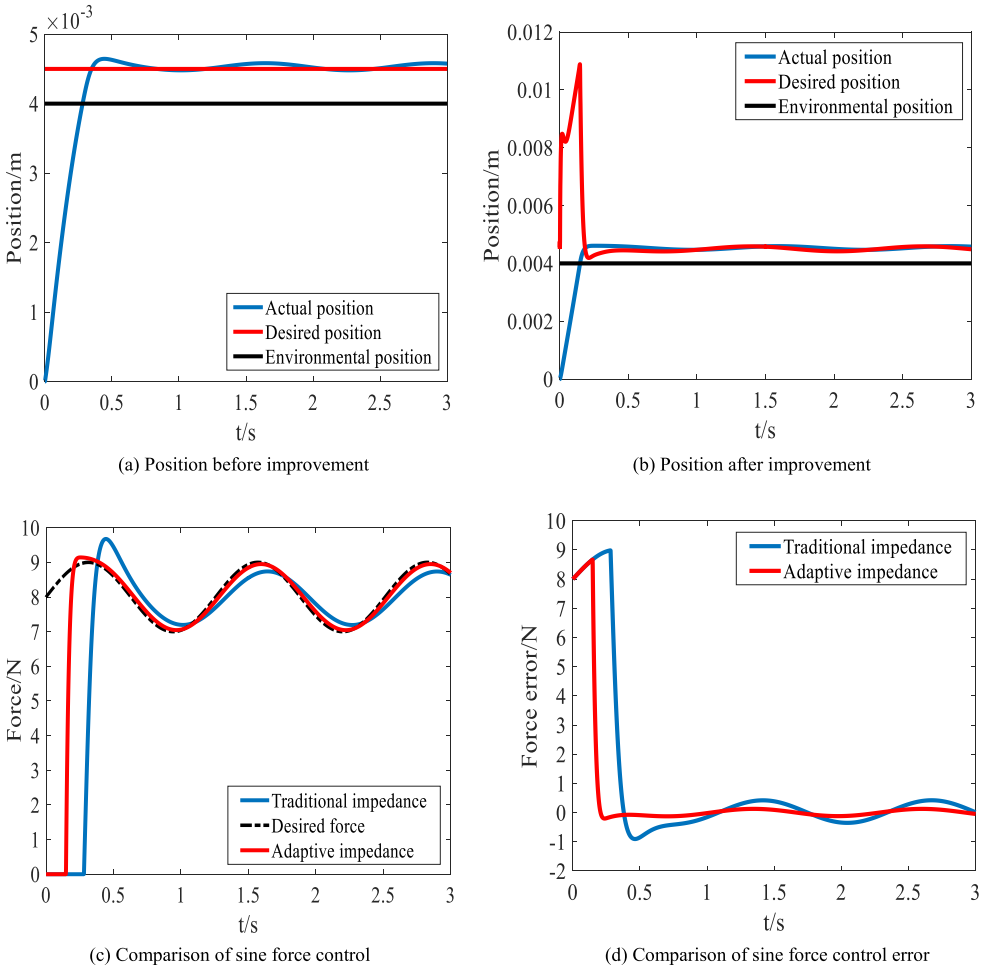

(a) Position before improvement

(b) Position after improvement

(c) Comparison of step force control

(d) Comparison of step force control error

**Fig. 9.** Step force tracking experiment under fixed environmental stiffness and environmental position.




(a) Position before improvement

(b) Position after improvement

(c) Comparison of sine force control

(d) Comparison of sine force control error

**Fig. 10.** Sine force tracking experiment under fixed environmental stiffness and environmental position.

realize stable and compliant grasping in the process of apple picking, combined with the above analysis about the minimum plastic deformation, it is determined that the third grasping mode causes the least mechanical damage, which is the optimal way to grasp apples.

### *3.2. Analysis of simulation results for adaptive impedance control system*

In order to make the designed controller appear good control characteristics, we referred to the controller parameter selection principle of Duan (Duan et al., 2018), and the specific parameters of the adaptive impedance controller were provided in Table 1.

#### *3.2.1. Simulation analysis under the condition of fixed environmental stiffness and environmental position*

The simulation time of the whole control system was set to 3 s, and the parameters of the controller system were set according to Table 1, we could obtain the step force and sine force tracking experimental results under the fixed environment stiffness and environmental position which were reflected in Fig. 9 and Fig. 10 respectively.

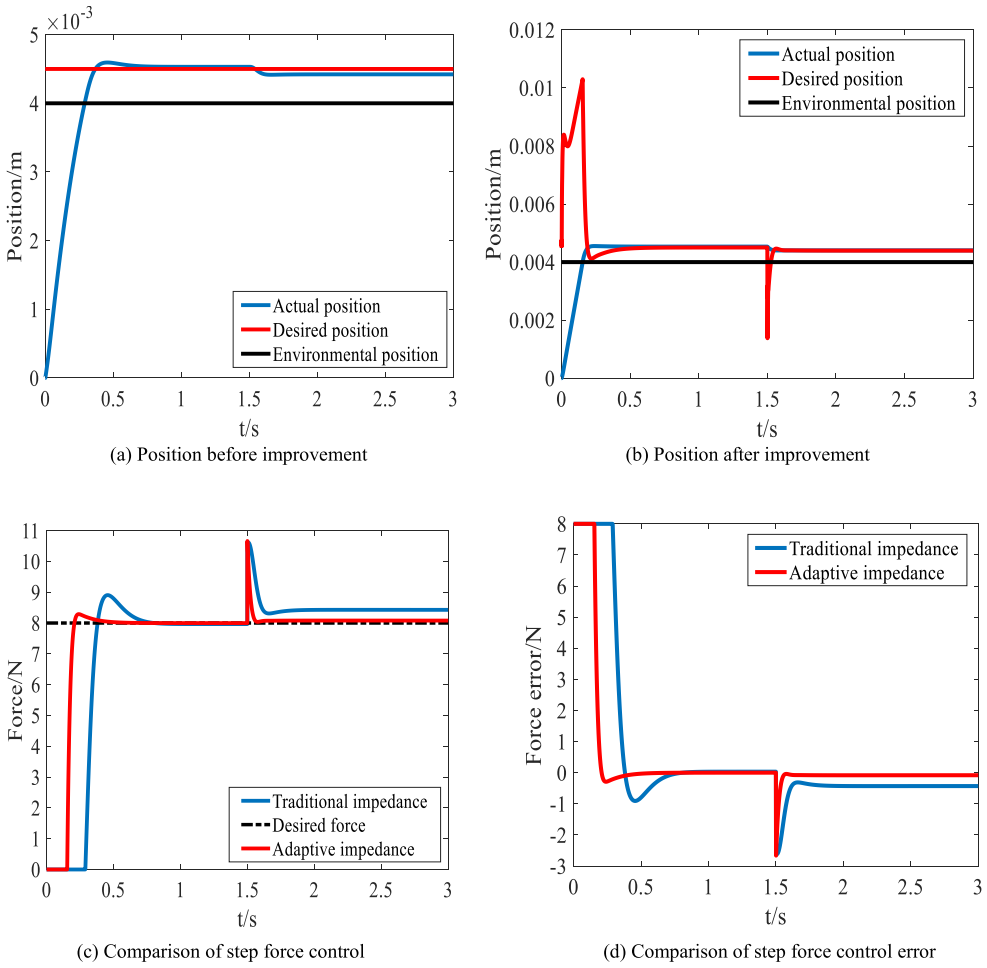
According to Fig. 9 that both control methods for tracking step force could be realized under the condition of the fixed environmental information. Compared with traditional impedance controller, the adaptive impedance controller tracked the desired force faster and its overshoot was significantly smaller. For the tracking of sine force in Fig. 10, grasping force of traditional impedance control presented a certain delay characteristic and it was difficult to enter the steady state, which may cause certain damage to apple. While that of adaptive impedance control showed better control effects in stability. More detailed data provided in Table 2 were used to compare the performance of controllers.

By analyzing the data in Table 2, the grasping force of adaptive impedance control reached the peak value of 8.29 N at about 0.24 s, its overshoot was about 3.6%. While that of traditional impedance control reached the peak value of 8.91 N at 0.46 s and its overshoot was about 11.38%. For sine force tracking, after the adaptive impedance control force reaching the peak value of 9.14 N at 0.25 s and the traditional impedance control force reaching the peak value of 9.67 N at 0.45 s, the former quickly adjusted the grasping force to a stable state around 0.34 s, while the latter did not

Aimed at the force tracking results in above figures and the

**Table 2**
Comparison between two controllers under fixed environmental stiffness and environmental position.

| | Step force | | | | Sine force | | |
|---|---|---|---|---|---|---|---|
| Control parameter | Peak force/N | Peak time/s | Overshoot/% | Adjustment time/s | Peak force/N | Peak time/s | Adjustment time/s |
| Traditional impedance | 8.91 | 0.46 | 11.38 | 0.66 | 9.67 | 0.45 | none |
| Adaptive impedance | 8.29 | 0.24 | 3.6 | 0.32 | 9.14 | 0.25 | 0.34 |




(a) Position before improvement

(b) Position after improvement

(c) Comparison of step force control

(d) Comparison of step force control error

**Fig. 11.** Step force tracking experiment under variable environmental stiffness.

experimental data in the table, we find that the adaptive impedance controller shows better control effect in terms of adjusting time and overshoot. When the gripper has not touched the apple, the contact force is 0, but compared with the traditional impedance control system, the adaptive impedance control system generates a position adjustment as the input of the position controller according to the error between the grasping force and the desired force, so the actual position of the gripper changes firstly and reaches the best grasping position because of the position adjustment, as a result, the actual grasping force of the adaptive impedance control reaches the desired force about 0.3 s before than that of the traditional impedance control, and the overshoot of the adaptive impedance control is smaller than that of the traditional impedance control.

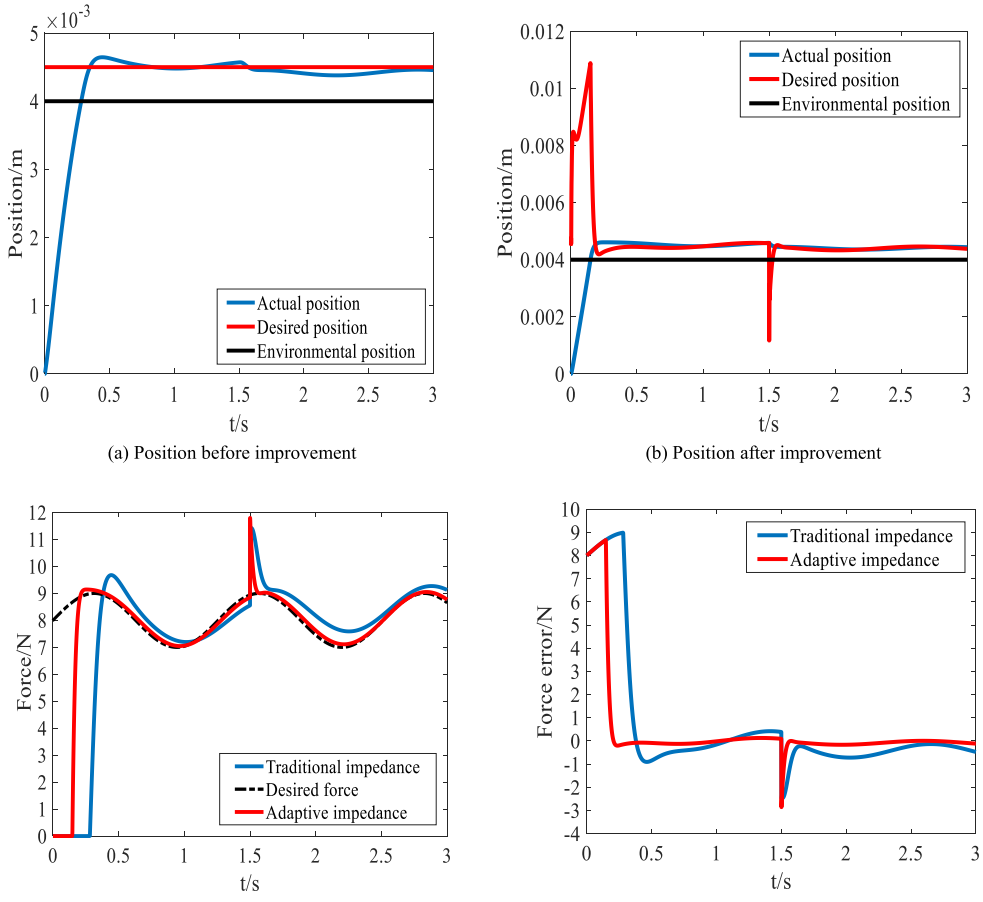
### *3.2.2. Simulation analysis under the condition of variable environmental stiffness*

In experiment 2, the other parameters of the control system remained unchanged except that the environmental stiffness was changed from 15,000 to 20,000 at 1.5 s, we could observe the control effect of step force and sine force tracking from the curves shown in Fig. 11 and Fig. 12 respectively.

As shown in Fig. 11, when the environmental stiffness changed suddenly in 1.5 s, the actual grasping force of the gripper changed sharply. Although the traditional impedance control system had a considerable adjustment effect on the grasping force, the final grasping force was still greater than the desired force. While that of adaptive impedance control tracked the desired force and the control system reached a stable state. For the tracking of sine force reflected in Fig. 12, the advantage of adaptive impedance control was more obvious. The anti-interference ability of the whole control system had been greatly enhanced in the case of ensuring the overshoot and adjustment time in an acceptable range. As a final point, the specific data about the force values in the simulation experiments were presented in the Table 3.

It can be seen from the data in the Table 3 that in the step force tracking experiment, when the environmental stiffness changed suddenly at 1.5 s, the grasping force of the adaptive impedance control system increased from 8 N to 10.66 N. In about 1.54 s, it was adjusted to a stable state and finally stabilized at 8.08 N. However, the grasping force of traditional impedance control system was only reduced from 10.62 N to 8.43 N after adjustment, which was far from the desired control requirements. In the process of tracking sine force, the mutation force of adaptive impedance control was 11.80 N, while that of traditional impedance control was 11.42 N. Although the former was larger than the latter, the former had better regulation effect than the latter.

In this simulation experiment, we simulate the variation of environmental stiffness as the variation of biological characteristics of apple in the process of grasping. Combined with Fig. 11, Fig. 12 and Table 3, we can discover that when the environmental stiffness changes, the actual grasping force of the gripper increases. At this time, the adaptive impedance control system begins to play a role. According to the large force error between the actual grasping force and the desired force, it produces a large amount of position adjustment, which makes the actual position of the gripper smaller and reach the desired position, and then the actual grasping force is adjusted to the desired value.




(a) Position before improvement

(b) Position after improvement

(c) Comparison of sine force control

(d) Comparison of sine force control error

**Fig. 12.** Sine force tracking experiment under variable environmental stiffness.

**Table 3**
Comparison between two controllers under variable environmental stiffness.

| | Step force | | | Sine force | |
|---|---|---|---|---|---|
| Control parameter | Mutation force/N | Final force/N | Adjustment time/s | Mutation force/N | Adjustment time/s |
| Traditional impedance | 10.62 | 8.43 | none | 11.42 | none |
| Adaptive impedance | 10.66 | 8.08 | 1.54 | 11.80 | 1.57 |

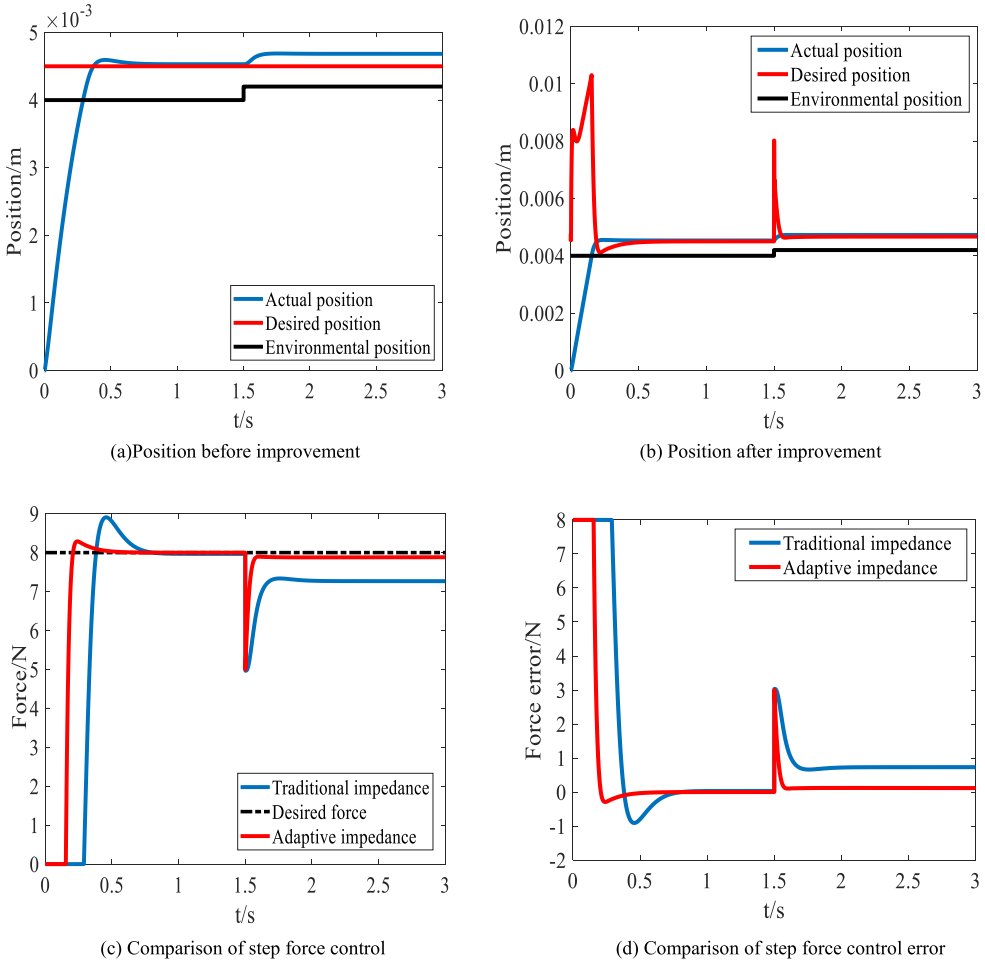
### *3.2.3. Simulation analysis under the condition of variable environmental position*

In experiment 3, considering that the bad selection of grasping point would change the environment position, the environment position was increased from 0.004 m to 0.0042 m at 1.5 s, while other parameters of the control system remained unchanged. The law of step force tracking and sine force tracking could be obtained through Fig. 13 and Fig. 14 respectively.

Due to the large value of environmental stiffness, the final grasping force had strong variation even if the environmental position changed little at 1.5 s. Aimed at the control effect of the step force tracking in Fig. 13, the traditional impedance control system showed poor robustness and the final grasping force was about 7.3 N, Although the adaptive impedance control system did not completely control the grasping force to the desired force value, it could quickly control the grasping force in a stable range, about 7.8 N. For the control effect of sine force tracking in Fig. 14, the adaptive impedance control system showed similar advantages in terms of adjusting time and overshoot as in the step force tracking experiment. The specific experimental comparison data were shown in Table 4.

The data in Table 4 reflect that when the environmental position became larger at 1.5 s in the process of tracking step force, the grasping force of the adaptive impedance control system was reduced from the original 8 N to 4.99 N, while that of the traditional impedance control system was reduced to 4.96 N. The former was adjusted by the adaptive impedance controller, and the final grasping force was stable at 7.8 N in 1.58 s, while the latter reached only 7.3 N, which was not in line with the requirements of force control. For sine force tracking, the grasping force of adaptive impedance control system and traditional impedance control system changed to 5.85 N and 5.55 N respectively, the former adjusted the grasping force to a small error range of the desired force in about 1.61 s, while the latter had a big error with the desired force.

In this experiment, considering that the improper selection of the grasping point will lead to the phenomenon of gripper sliding when grasping the apple, which will change the environmental position, resulting in large error between the actual grasping force and desired force. We simulated the situation where the environmental position changed. From the above experimental results, when the environmental position changes, the force error becomes larger visible to the naked eye due to the large environmental stiffness. At this time, the adaptive




(a)Position before improvement

(b) Position after improvement

(c) Comparison of step force control

(d) Comparison of step force control error

**Fig. 13.** Step force tracking experiment under variable environmental position.

impedance control system increases the actual position value of the gripper according to the force error value until the actual grasping force value reaches the acceptable stable range. However, the traditional impedance control system lacks the adjustment of this step, resulting in a large error between the actual grasping force and desired force.

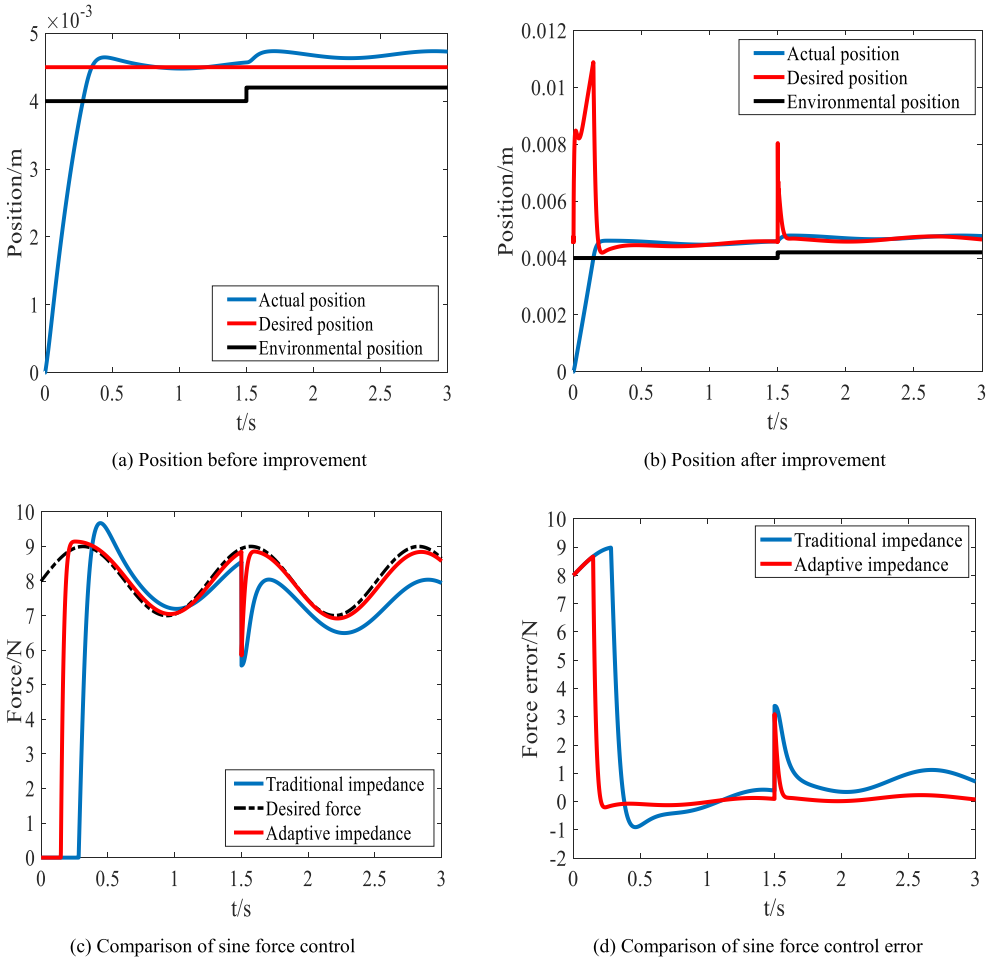
According to the three experiments mentioned above, it can be concluded that the adaptive impedance control strategy proposed in this paper has strong robustness and compensation for the sudden change of the environmental stiffness and the environmental position, and it can quickly adjust the desired position so that the contact force can track the desired force with low overshoot.

### 3.3. Experiment results analysis

In order to verify the effectiveness of the proposed algorithm, the traditional impedance control strategy and adaptive impedance control strategy were used to carry out the comparative experiment of apple grasping, respectively. The curve of grasping force with grasping time could be obtained, as shown in Fig. 15. It could be seen from the figure that in the stage of free movement, the contact force was 0 because of that the gripper did not touch the apple. When the gripper touched the apple, these two control strategies gave full play to the effect to adjust the output torque of the motor so that the grasping force could quickly track the desired force. And when the gripper grasped the apple stably, the IPC would send a signal to the cutting device to cut off the stem. Finally, the gripper would release the apple at designated position.

By comparing the experimental result, we could find that the adaptive impedance control reached the peak value of 8.25 N at about 0.24 s while the traditional impedance control reached the peak value of 8.58 N at 0.47 s respectively. The adaptive impedance control not only had some advantages in adjusting time, but also significantly improved its overshoot. The overshoot of the adaptive impedance control was 3.125% and the overshoot of the traditional impedance control was 7.25%. At the same time, the grasping force of adaptive impedance control varied from7.86 N to 8.09 N, while that of traditional impedance control fluctuated from 7.76 N to 8.35 N. The adaptive impedance control strategy was further confirmed to be more compliant and suitable for apple harvesting robot compared with the traditional impedance control strategy, which was consistent with the simulation results.

According to the above simulation results and the experimental results, the compliant grasping of apple is completed, which also proves that the adaptive impedance control algorithm proposed in this paper plays a certain role. The fundamental reason is that when the actual grasping force of the gripper is different from the desired force, the adaptive control algorithm begins to work. According to the magnitude of force error, the adaptive controller outputs an appropriate position adjustment, which adjusts the actual position of the gripper together with the output of the impedance controller, so that it can reach the desired position faster than the traditional impedance control system. At the same time, the actual grasping force of the gripper is adjusted to the desired force faster along with the variation of the actual position. The results of this study can provide important guiding significance for the future robotic system of grasping fruit which are vulnerable to injury and other occasions with high control accuracy, it can help robotic system to improve the success rate of grasping and reduce the damage to the target fruit as much as possible.




(a) Position before improvement

(b) Position after improvement

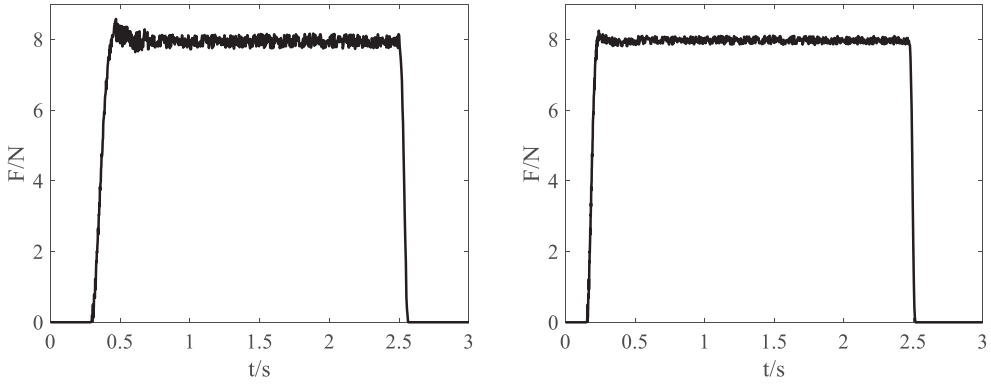
(c) Comparison of sine force control

(d) Comparison of sine force control error

**Fig. 14.** Sine force tracking experiment under variable environmental position.

**Table 4**
Comparison between two controllers under variable environmental position.

| | Step force | | | Sine force | |
|---|---|---|---|---|---|
| Control parameter | Mutation force/N | Final force/N | Adjustment time/s | Mutation force/N | Adjustment time/s |
| Traditional impedance | 4.96 | 7.3 | none | 5.55 | none |
| Adaptive impedance | 4.99 | 7.8 | 1.58 | 5.85 | 1.61 |




(a) Grasping force before improved

(b) Grasping force after improved

**Fig. 15.** Comparison of experimental grasping force.

## 4. Conclusions

The four element Burgers model is used to characterize the rheological properties of apples, and the viscoelastic parameters of the apple are obtained through a compression test. The stress analysis of each viscoelastic layer is carried out to summarize the relationship between apple contact force and displacement of the gripper, then the model of apple plastic deformation is proposed. Combined with the measured viscoelastic parameters of apples, the minimum plastic deformation of the apple under three different grasping modes is calculated and the optimal grasping mode is determined.

In order to realize compliant picking, the position-based impedance control system is established. Then aiming at the influence of sudden changes in environmental stiffness and environmental position, adaptive theory and Lyapunov stability theorem II are used to design an adaptive controller to improve the traditional impedance control system by adding a position adjustment to the desired position. The control effect of the situation of fixed environmental stiffness and environmental position and the situation of variable environmental stiffness and environmental position are compared by simulation analysis. In the case of no sudden change of environmental information, the force overshoot of traditional impedance control for step force is 11.38%, and then it can converge to the stable state, while the adjustment time of the adaptive impedance control system is shorter, and the overshoot is only 3.6%. When there is a sudden change in environmental information, although traditional impedance control system shows a certain degree of robustness, the grasping force has not reached a steady state. While the adaptive impedance control system makes the grasping force reach a steady state, and its control performance and the anti-interference ability have been greatly improved compared with the traditional impedance control system. The final experimental results show that the proposed control strategy is consistent with the simulation results.

## CRediT authorship contribution statement

**Wei Ji:** Conceptualization, Methodology, Writing - review & editing, Project administration, Funding acquisition. **Jiangwei Zhang:** Data curation, Validation, Writing - review & editing, Formal analysis. **Bo Xu:** Visualization, Investigation, Funding acquisition. **Chencheng Tang:** Methodology, Software, Resources, Writing - original draft. **Dean Zhao:** Resources, Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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